

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 40-2021: A 9-Year-Old Boy with Transient Weakness, Facial Droop, and Slurred Speech

Miya E. Bernson-Leung, M.D., Ed.M., Otto Rapalino, M.D.,
Melissa A. Walker, M.D., Ph.D., and William E. Butler, M.D.

PRESENTATION OF CASE

Dr. Sagar M. Raju (Pediatrics): A 9-year-old boy was transferred to this hospital because of transient weakness and facial droop on the left side and slurred speech.

The patient had been well until 5 hours before this evaluation, when his parents observed that he had a facial droop on the left side and slurred speech. In addition, the patient noticed that his left arm and leg felt “limp.” When he moved to get out of his bed, he could not stand on his own or grip using the left hand. There was no confusion, loss of consciousness, shaking in the arms or legs, urinary or fecal incontinence, headache, nausea, vomiting, recent illness, or trauma. The symptoms lasted approximately 2 minutes and then resolved completely. Emergency medical services were called, and the patient was taken to the emergency department of another hospital.

At the other hospital, the physical examination was reportedly normal. The complete blood count, erythrocyte sedimentation rate, prothrombin time, partial-thromboplastin time, and blood levels of electrolytes, glucose, and troponin were normal, as were the results of tests of renal, liver, and thyroid function. The patient was transferred to this hospital for further evaluation.

On arrival at this hospital, additional history was obtained from the patient’s parents. Earlier in the day, the patient had become upset while visiting his grandfather, who was dying of esophageal cancer. He then played a game of hockey with a sibling; no falls or head trauma occurred. At the time that his parents observed the new neurologic symptoms, he had been lying in bed and crying profusely and inconsolably for an extended period. His parents noticed that he was gasping for air and hiccupping while crying.

The patient had been born by normal spontaneous vaginal delivery at full term. He had met normal developmental milestones and had received routine childhood vaccinations. He had chronic gastrointestinal upset that had been attributed to anxiety. Two years earlier, he had been evaluated at the neurology clinic of another hospital because of episodes of shaking in the arms and legs, which occurred multiple times a day. The shaking occurred more frequently when the patient was excited or emotional. He was aware of the shaking; he felt an urge before each episode, and the urge was relieved after the shaking occurred. His parents

From the Department of Neurology, Boston Children’s Hospital (M.E.B.-L.), the Departments of Neurology (M.E.B.-L., M.A.W.), Radiology (O.R.), and Neurosurgery (W.E.B.), Harvard Medical School, and the Departments of Radiology (O.R.), Neurology (M.A.W.), and Neurosurgery (W.E.B.), Massachusetts General Hospital — all in Boston.

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were told that the shaking was stereotypy, and no diagnostic tests were performed.

The patient took no medications and had no known drug allergies. He lived with his parents and two siblings in a suburban area of New England. He attended elementary school and performed well academically, although he received tutoring for mild difficulties with reading and phonology. In addition to the history of esophageal cancer in his paternal grandfather, his father had ulcerative colitis, his mother had received a diagnosis of large-B-cell lymphoma in childhood, his sister had asthma, and his maternal grandfather had died of acute coronary syndrome at 30 years of age.

On evaluation, the patient appeared well developed and mildly anxious. The temperature was 37.2°C, the blood pressure 116/64 mm Hg, the pulse 91 beats per minute, the respiratory rate 26 breaths per minute, and the oxygen saturation 100% while he was breathing ambient air. The face was symmetric, with normal forehead wrinkling, blinking, smiling, and cheek puffing. An evaluation of other cranial-nerve functions, sensation, power, bulk, tone, coordination, and gait was normal.

A diagnostic test result was reviewed, and a diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Miya E. Bernson-Leung: In a 9-year-old boy with transient neurologic symptoms, the differential diagnosis is broad, encompassing relatively benign to potentially life-threatening entities. Even symptoms that have completely resolved by the time of presentation may be harbingers of a serious condition. Because there were no observable signs when this patient presented to this hospital, obtaining a careful history is crucial to the diagnostic reasoning process. To guide further evaluation and treatment, the clinician must simultaneously determine the type of disease process that may be occurring and the area in the nervous system that may be involved.

NEUROLOGIC SYMPTOMS

A detailed description of the neurologic event (known as the semiology) should be obtained from the patient, his parents, and any other observers. Useful descriptors include the duration and number of episodes; the tempo of onset and

offset (abrupt or gradual); the evolution or sequence of symptoms; the similarity of symptoms across episodes (variable or stereotyped), if applicable; any prodromal or postdromal symptoms, including both neurologic and systemic symptoms; and any provoking or ameliorating factors. It is important to understand specifically what the patient or observer means when using terms such as pain, weakness, numbness, or dizziness. Finally, if the event has been recorded, review of the video can be invaluable.

On the basis of the event description, the neurologic symptoms can be characterized as either focal or diffuse. This characterization may provide clues to the area involved. Focal symptoms may point to a problem in a specific portion of the neuraxis. Diffuse symptoms, such as altered mental status, headache, or generalized weakness, are not as readily localized. Characterization of the symptoms as focal or diffuse may also provide clues to the type of disease process. There are conditions that may cause focal symptoms (e.g., trauma or vascular disease), conditions that tend to cause diffuse symptoms (e.g., toxic-metabolic encephalopathy, systemic infection, or inflammation), and conditions that can cause both (e.g., intracranial infection or some mass lesions).

It is also useful to characterize the neurologic symptoms as positive or negative. Excitatory processes such as seizure or migraine with aura tend to cause positive symptoms (or excesses), such as involuntary movements, olfactory hallucinations, paresthesias, or the scintillations and other visual phenomena of migraine with aura. Inhibitory processes such as demyelination or ischemia tend to produce negative symptoms (or deficits), such as weakness, dyscoordination, numbness (hypoesthesia or anesthesia), visual-field cuts, or loss of color perception. There are many exceptions to this broad rule, including postictal paralysis (Todd's paralysis) and hemiplegic migraine with aura; in these cases and others, the history may indicate a mix of positive and negative symptoms.

This patient's history is notable for the development, when he was 7 years of age, of episodes of shaking in the arms and legs. The episodes occurred repeatedly throughout the day in a stereotyped fashion, were associated with a prodromal urge and postdromal relief, and were exacerbated by excitement or strong emotion. These symp-

toms are focal and positive. They are consistent with motor tic disorder, a childhood movement disorder that is common, benign, and usually self-limited and can be diagnosed clinically.

By contrast, the symptoms described during this patient's current presentation were hemibody weakness (hemiparesis) and slurred speech (dysarthria), which had a sudden onset, were preceded by crying and emotional upset, and resolved completely after 2 minutes without intervention. These symptoms are focal and negative. The acute onset of focal neurologic deficits describes a strokelike episode, or brain attack.¹

BRAIN ATTACK

Some brain attacks are caused by ischemic stroke or transient ischemic attack. However, similar presentations of acute focal neurologic deficits can have other causes, which can be broadly grouped into the following categories: electrochemical, pressure-mediated (including hemorrhagic stroke), toxic or metabolic, musculoskeletal, infectious or inflammatory, and somatoform (Fig. 1). Many brain attacks are transient or fluctuating but can still progress to permanent injury. Also, many causes of brain attack are urgent or emergency conditions that require prompt treatment to prevent or minimize neurologic disability and other complications.¹ These concerns have led to the development of brain-attack or stroke-alert rapid response teams at many pediatric centers.

In one series of 124 pediatric stroke-alert activations for acute focal neurologic deficits,² 30 (24%) were associated with a final diagnosis of stroke and 2 (2%) with a final diagnosis of transient ischemic attack. Of the 21 patients with ischemic stroke, 3 were eligible for and underwent targeted therapy for hyperacute stroke (intravenous administration of tissue plasminogen activator or endovascular thrombectomy). An additional 17 patients (14%) had other neurologic emergencies, a finding that highlights the importance of the rapid response pathway.

This patient had several clinical features suggestive of ischemic stroke or transient ischemic attack. First, the sudden onset of focal deficits is the hallmark feature of both stroke and transient ischemic attack, although progressive or fluctuating symptoms do not rule out these diagnoses, especially in younger patients.^{3,4} Second, there was no preceding convulsive activity sug-

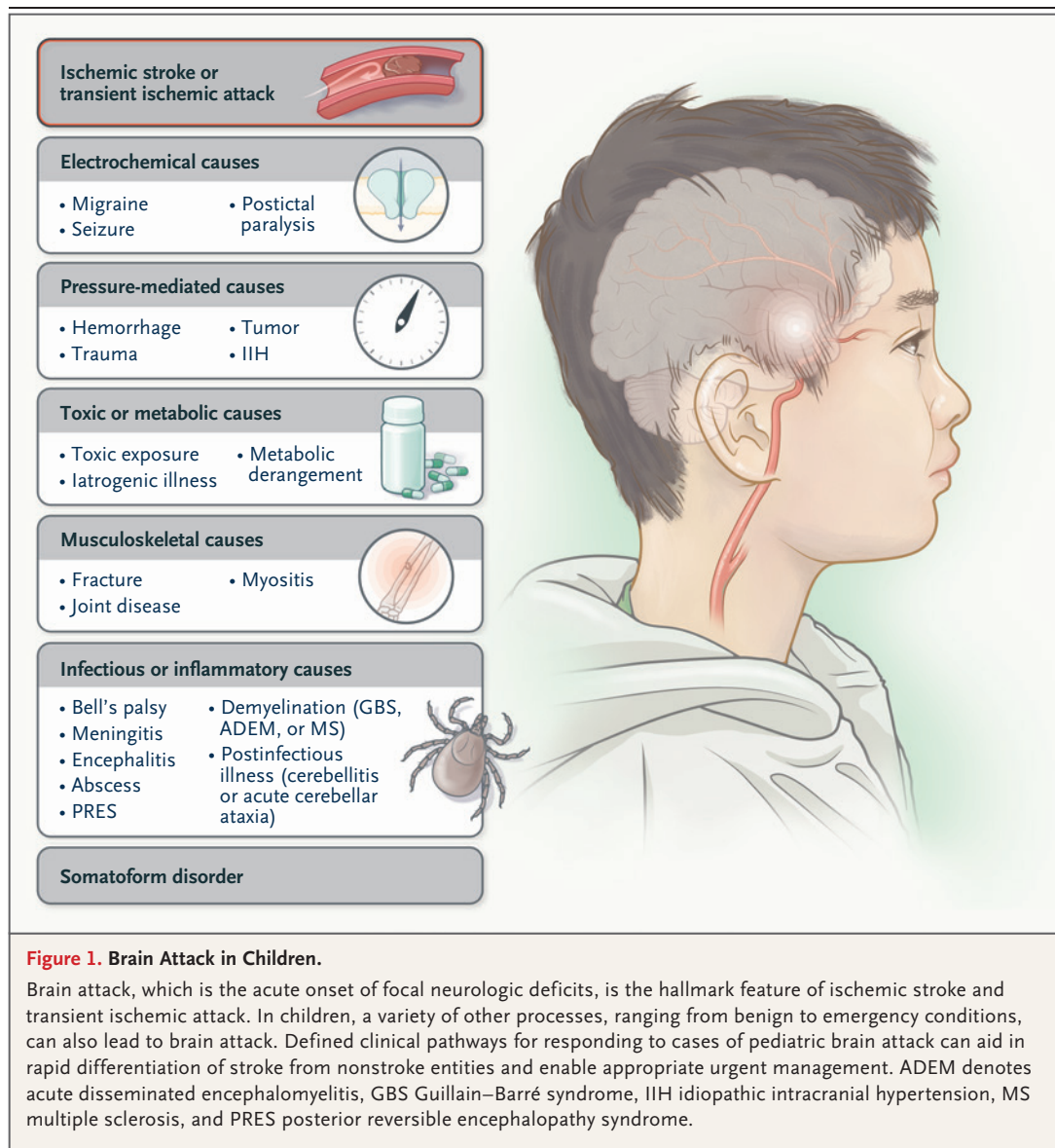
gestive of postictal paralysis. Third, the deficits could be localized to a single vascular territory.

Arterial ischemic stroke is defined as the blockage of a cerebral blood vessel that leads to ischemia, and ultimately to irreversible infarction, of the region of brain tissue supplied by that vessel. Transient ischemic attack involves the same process, but the process is reversed before progression to infarction, with complete resolution of symptoms. In either case, the constellation of deficits corresponds to the functions of the ischemic area (or areas). This patient had weakness of the face, arm, and leg on the left side, with dysarthria due to motor weakness of the face but without aphasia. This pure motor syndrome involves weakness of the entire hemibody but spares language and visuospatial functions, pointing to ischemia of deep structures such as the internal capsule and adjacent basal ganglia, where the descending motor pathways are closely apposed. The affected region is supplied by small perforating branches off the middle cerebral arteries.

Arterial ischemic stroke results from one or more of three mechanisms (Fig. 2): thrombosis, stenosis, and thromboembolism. In thrombosis, there is an increased tendency toward the formation of blood clots. In stenosis, a cerebral artery becomes narrowed because of arterial dissection or intrinsic abnormalities of the vasculature (e.g., moyamoya arteriopathy, sickle cell vasculopathy, or other arteriopathies), which can lead to downstream hypoperfusion. In thromboembolism, clots formed elsewhere in the body (a more proximal artery, the heart, or the venous circulation in the case of paradoxical embolism through an intracardiac shunt) move and become lodged in a cerebral artery. In children, these three mechanisms can have a variety of causes, including congenital or acquired arteriopathies (inflammatory, infectious, traumatic, and other arteriopathies), cardiac disease, regional or systemic infection and inflammation, and genetic or acquired thrombophilic conditions. In many cases, two or more associated factors can be identified at the time of stroke.⁵

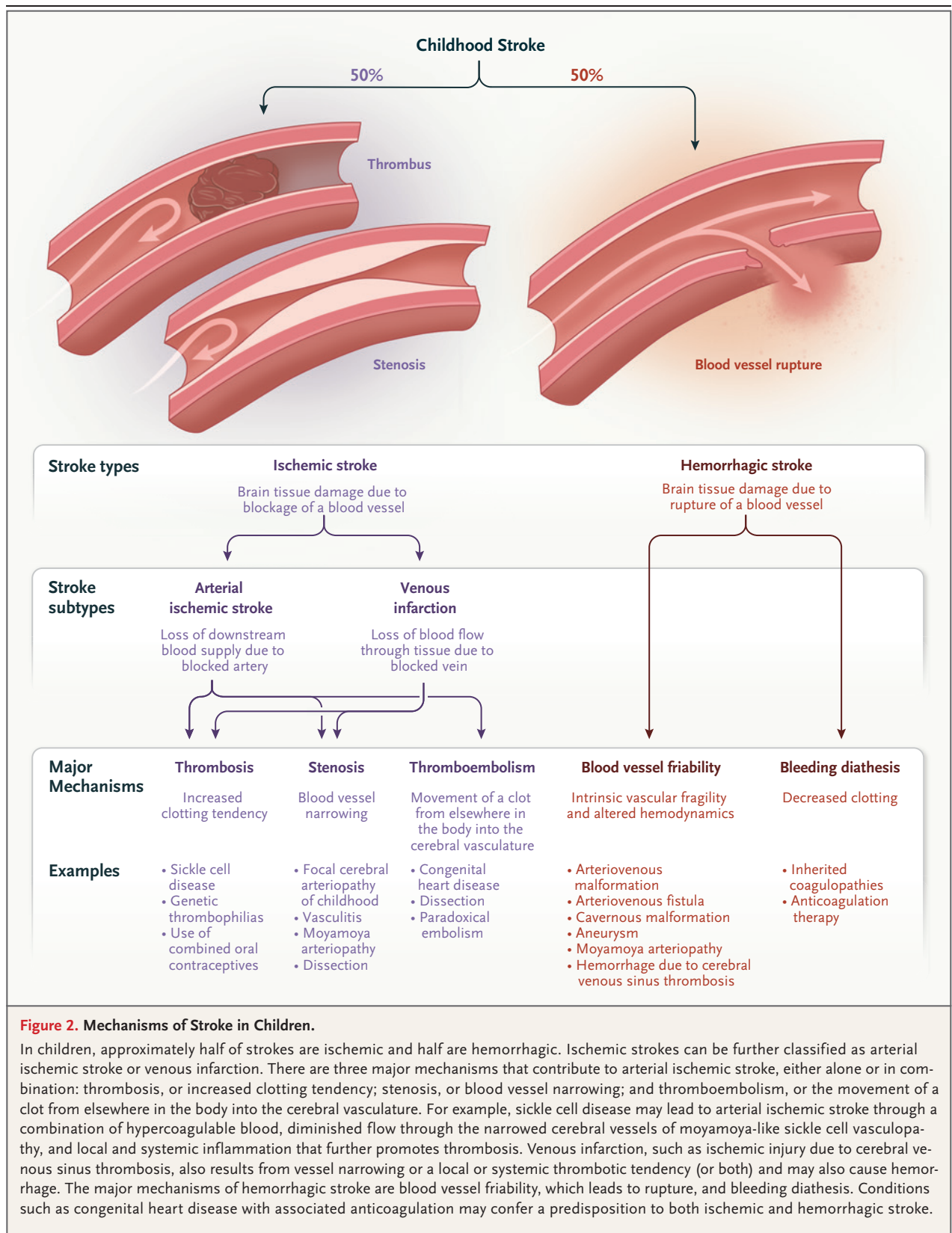
NEUROIMAGING

The definitive test to establish the diagnosis of acute ischemic stroke or transient ischemic attack and to begin the assessment for underlying



causes is neuroimaging with angiography of the head and neck. In children, the preferred technique is magnetic resonance imaging (MRI) of the head with magnetic resonance angiography (MRA) of the head and neck. This approach can achieve the three goals of imaging for hyperacute stroke while avoiding exposure to radiation and contrast material. First, diffusion-weighted MRI can show evidence of ischemia within minutes of onset, whereas computed tomography (CT) may show no abnormalities in the early phase of stroke. Second, perfusion-weighted MRI and MRA can be used to identify vascular occlusions or abnormalities that confirm the diagnosis of

stroke, may reveal an arteriopathy as an underlying cause, and may render the patient eligible for clot-directed therapies such as the intravenous administration of tissue plasminogen activator or thrombectomy. Third, susceptibility-weighted or gradient-echo MRI can be used to rule out hemorrhage; hemorrhagic stroke is high on the differential diagnosis of ischemic stroke and would require different management strategies.⁴ Even when the symptoms suspicious for transient ischemic attack have resolved, it is important to perform MRI with MRA to assess for ischemic injury (given that some pediatric strokes apparent on imaging have resolving symptoms)



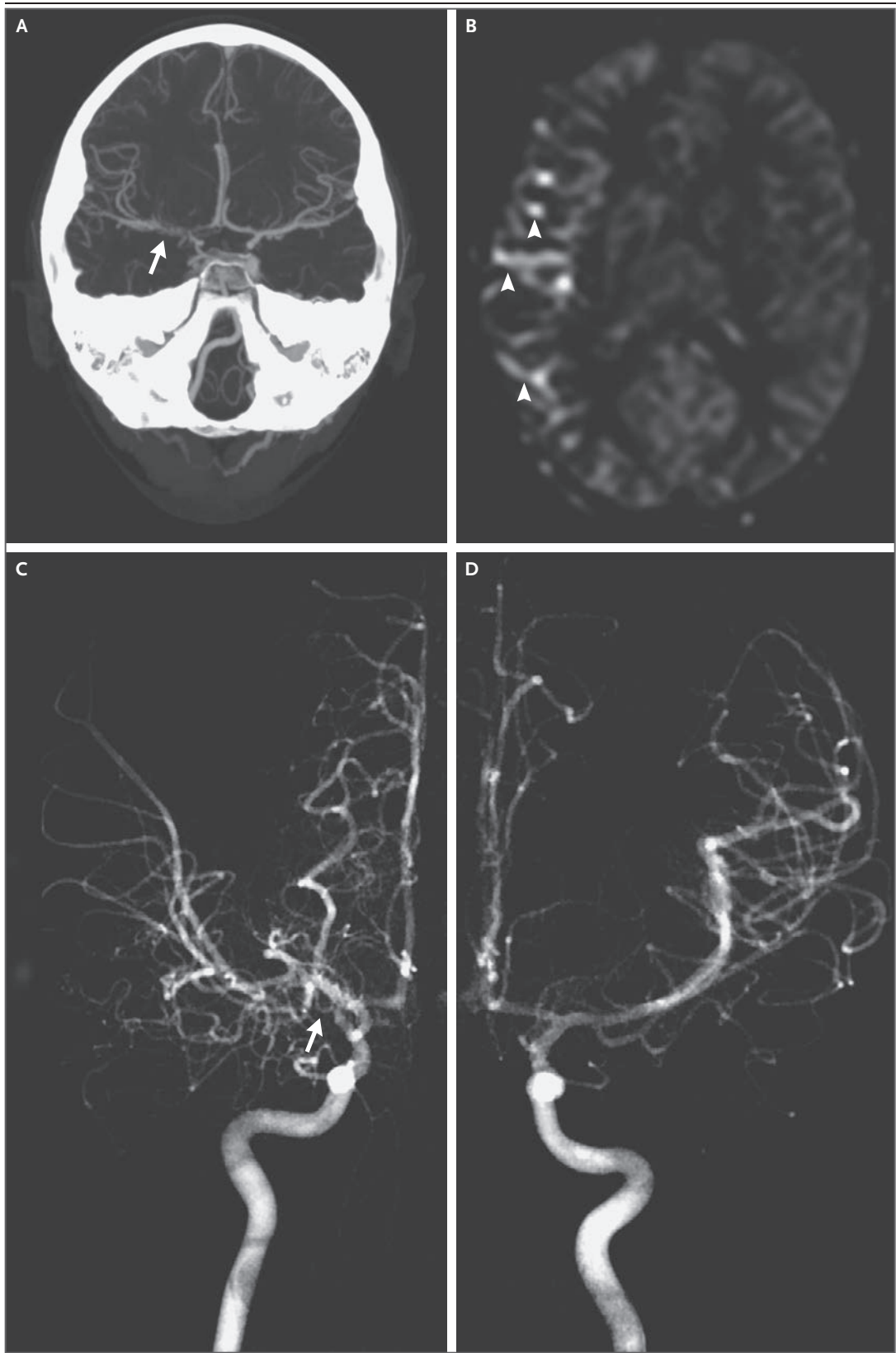


Figure 3 (facing page). Initial Imaging Studies.

Shown are a coronal maximum-intensity-projection (MIP) reformatted image of the initial CT angiogram (Panel A), an axial pulsed arterial spin labeling (PASIL) perfusion-weighted MRI (Panel B), and conventional frontal digital subtraction angiograms obtained during injection of contrast material into the right and left internal carotid arteries (Panels C and D, respectively). The CT angiogram and a conventional angiogram show focal severe stenosis of the horizontal (M1) segment of the right middle cerebral artery (Panels A and C, arrows), which is surrounded by collateral vessels in a moyamoya pattern. The other conventional angiogram shows no abnormalities (Panel D). The PASIL image shows multiple foci of increased signal intensity along the distal branches of the right middle cerebral artery (Panel B, arrowheads), a finding that probably reflects an arterial transit delay artifact and focal decreased signal intensity in the right frontal white matter.

and to identify underlying causes that could lead to a future completed stroke.^{6,7}

MOYAMOYA ARTERIOPATHY

In this patient, the fact that the episode of neurologic deficits was preceded by crying is notable. Crying may lead to hyperventilation, which is a known trigger for transient ischemic attack and stroke in children with moyamoya arteriopathy. This condition is a progressive steno-occlusive disease of the major cerebral vessels, with compensatory hyperplasia of perforating branches and other collateral vessels and ultimately failure of cerebral perfusion in affected vascular territories.⁸ When moyamoya arteriopathy is idiopathic, it is called moyamoya disease. When it occurs in the context of known associated conditions, such as sickle cell disease, cranial irradiation, or Down syndrome, it is called moyamoya syndrome.

It is thought that the decreased partial pressure of carbon dioxide that occurs during hyperventilation leads to cerebral vasoconstriction, resulting in ischemia of areas with tenuous perfusion.⁹ This patient's hypertension at presentation could reflect systemic compensation to maximize cerebral perfusion. In returning briefly to the patient's history of possible motor tics beginning when he was 7 years of age, motor tics and other movement disorders, which are the hallmarks of basal ganglia abnormalities, have been described in patients with untreated moyamoya arteriopathy. They may resolve spontaneously or after revascularization procedures.¹⁰

This phenomenon may be due to the direct mass effect of collateral vessel proliferation on the basal ganglia or to abnormal basal ganglia perfusion even without infarction.^{11,12} In summary, this patient's history suggests that he had a transient ischemic attack, possibly due to moyamoya disease. To establish the diagnosis of moyamoya disease in this case, neuroimaging with angiography should be performed.

**DR. MIYA E. BERNSON-LEUNG'S
DIAGNOSIS**

Transient ischemic attack due to moyamoya disease.

DIAGNOSTIC TESTING

Dr. Otto Rapalino: The patient underwent CT angiography of the head, which revealed focal severe stenosis of the terminus of the right internal carotid artery and of the adjacent proximal (A1) segment of the right anterior cerebral artery. In addition, the horizontal (M1) segment of the right middle cerebral artery appeared to be severely narrowed and was surrounded by prominent collateral vessels (Fig. 3A). The remaining intracranial arteries were patent, without focal stenosis. There was no evidence of hemorrhage or acute infarction on the initial CT images.

Subsequent MRI of the head confirmed the absence of acute infarction and intracranial hemorrhage. Pulsed arterial spin labeling (PASIL) perfusion-weighted images showed asymmetrically increased signal intensity along the distal branches of the right middle cerebral artery, a finding that probably reflected an arterial transit delay artifact and decreased signal intensity in the right frontal white matter (Fig. 3B). MRA confirmed the presence of focal severe narrowing of the terminus of the right internal carotid artery and showed poor visualization of the adjacent right A1 and M1 segments, with replacement of the right M1 segment with prominent collateral vessels in a moyamoya pattern.

After MRI with MRA was performed, the patient underwent conventional transfemoral cerebral angiography to further characterize the vascular findings. Conventional angiography revealed severe stenosis of the right M1 segment, mild tapering of the terminus of the right internal

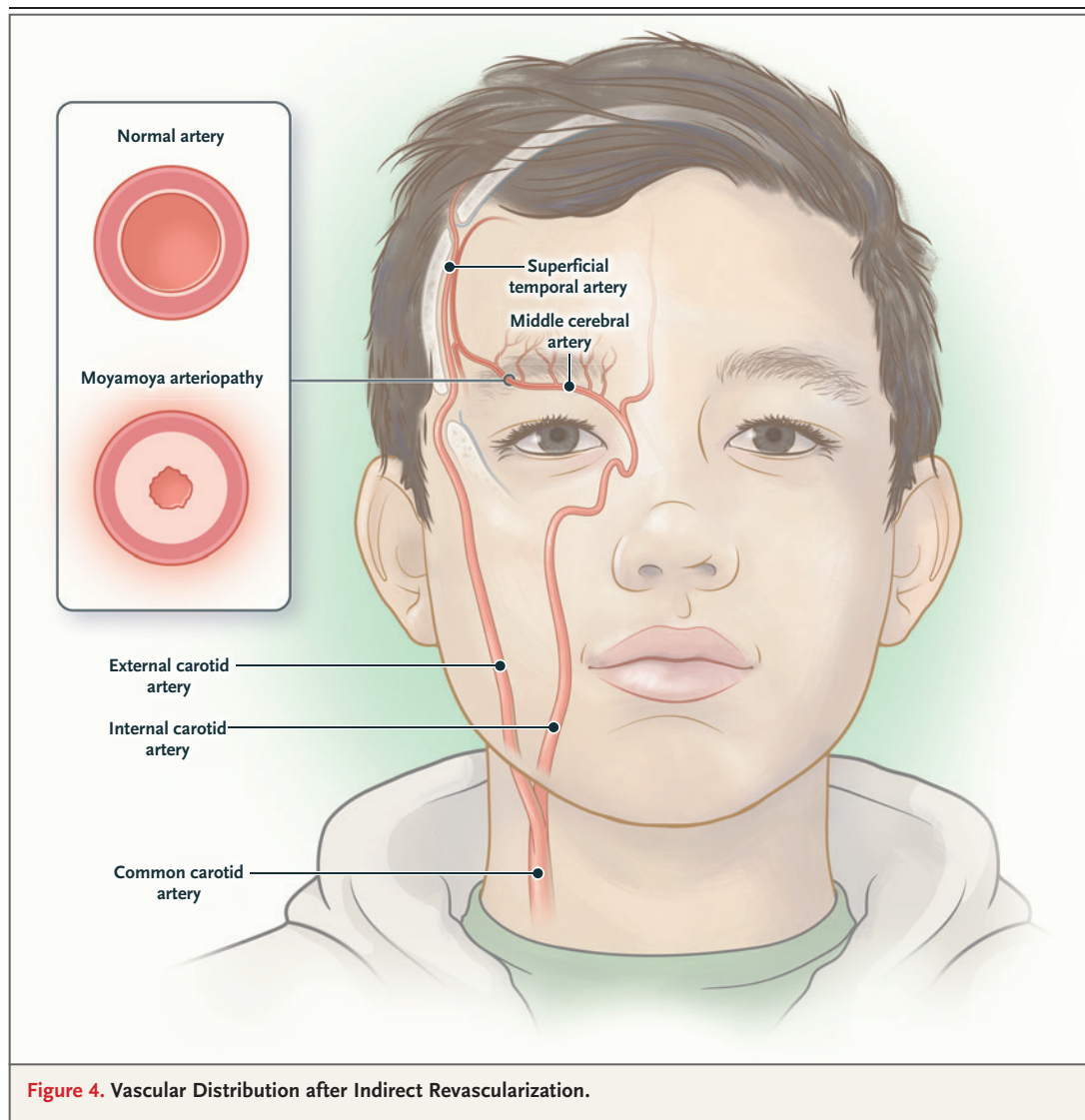


Figure 4. Vascular Distribution after Indirect Revascularization.

carotid artery, and a robust right superficial temporal artery (Fig. 3C).

RADIOLOGIC DIAGNOSIS

Moyamoya arteriopathy.

DISCUSSION OF MANAGEMENT

Dr. Melissa A. Walker: No therapies for moyamoya disease or syndrome have been approved by the Food and Drug Administration. Mainstays of medical management are aimed at prevention of secondary stroke, maintenance of cerebral perfusion, and treatment of associated headache and seizure. To promote cerebral perfusion, intravenous hydration may be provided, and blood

pressure-lowering agents should be avoided in the acute period of illness. It is important to minimize pain (e.g., with the use of painless wound-handling techniques) and stressors that could lead to crying and secondary hyperventilation, which could, in turn, result in hypocapnia and cerebral vasoconstriction.⁸ Patients with moyamoya arteriopathy may have coexisting conditions that could affect these considerations, such as sickle cell disease, which is associated with complex pain management, or Down syndrome, which requires adaptation to the psychosocial stressors of hospitalization. Expert consensus recommends the immediate administration of antiplatelet therapy with aspirin for the treatment of arterial ischemic stroke related to moyamoya arteriopathy in children and also for

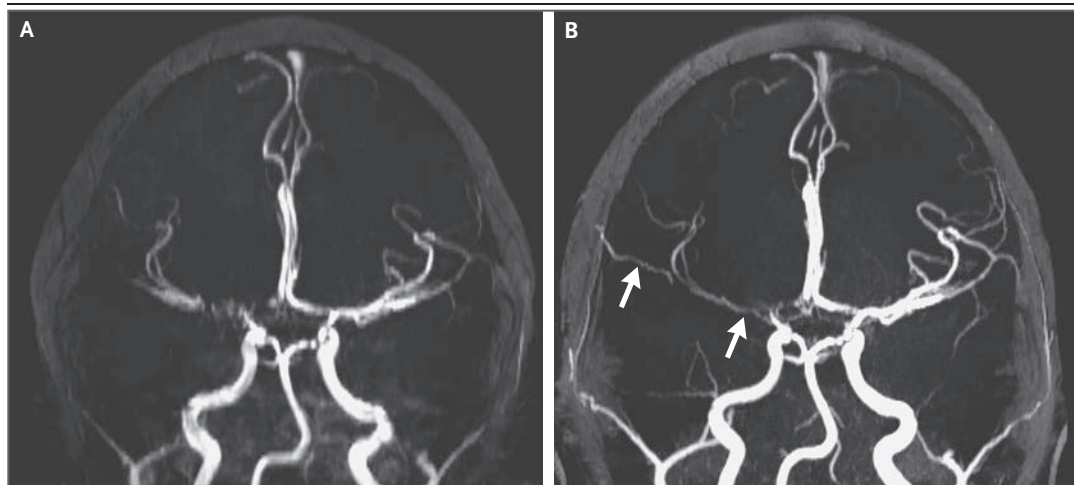


Figure 5. Preoperative and Postoperative Imaging Studies.

Shown are coronal MIP reformatted images of magnetic resonance angiograms obtained preoperatively and postoperatively (Panels A and B, respectively). The postoperative image shows a slight increase in the flow-related signal in the distal branches of the right middle cerebral artery that are near the surgical site and in the right M1 segment (Panel B, arrows).

secondary stroke prevention.¹³ Medical management should also include the administration of appropriate antiepileptic agents for seizure control and abortive and prophylactic therapies for headache. Important considerations for medication selection in patients with moyamoya arteriopathy include the need to avoid (if possible) vasoactive medications (e.g., calcium-channel blockers, beta-blockers, or triptans), as well as the potential for an increased risk of hemorrhage associated with indicated treatments (i.e., concurrent aspirin and other nonsteroidal anti-inflammatory drugs).¹⁴

DISCUSSION OF SURGICAL MANAGEMENT

Dr. William E. Butler: This patient's imaging studies showed almost total obstruction at the terminus of the right internal carotid artery in a moyamoya pattern and suggested contralateral stenosis. Diffusion-weighted MRI did not show evidence of acute or old infarction. PASL perfusion-weighted MRI showed increased transit times in the distribution of the right middle cerebral artery.¹⁵ This finding indicates a risk of eventual right cerebral infarction due to arterial insufficiency.

The patient was admitted to the pediatric intensive care unit for close clinical observation, continuous management of the arterial blood

pressure, maintenance of euolemia, and administration of aspirin therapy. During the hospitalization, he did not have any additional episodes of weakness on the left side.

Children with moyamoya arteriopathy who undergo surgical revascularization appear to have a lower long-term risk of cerebral infarction than those who receive medical therapy alone.^{16,17} There are indirect and direct revascularization techniques,^{18,19} and the establishment of general anesthesia is challenging.²⁰ Both techniques are associated with a risk of intraoperative stroke due to hypotension (Fig. 4).

With indirect revascularization, well-perfused extracranial tissues, including the superficial temporal artery, are placed directly onto underlying ischemic brain tissue to allow for the gradual development of neoangiogenic connections over the course of months. With direct revascularization, the indirect method is augmented by creation of a suture anastomosis, typically between the superficial temporal artery and a branch of the middle cerebral artery. The direct technique requires temporary clip occlusion, arteriotomy, and suturing of the superficial temporal artery and a branch of the middle cerebral artery; it immediately results in increased blood flow. The temporary clip occlusion is associated with risks of infarction during the period of occlusion, infarction due to thrombosis at the suture anastomosis site, and hemorrhage.

In this case, catheter-based cerebral angiography was performed for two reasons. The first reason was to determine whether stenosis of the contralateral internal carotid artery was present, which could lead to the performance of bilateral revascularization. The second reason was to inform the recommendation for either direct or indirect anastomosis.

Cerebral angiography revealed a patent left internal carotid artery. The right internal carotid artery showed delayed filling but sufficient flow to result in neoangiogenesis. In addition, the right superficial temporal artery had a diameter of 1 mm or less, which is unfavorable for serving as a donor vessel in a direct suture anastomosis. Therefore, we performed indirect revascularization.

Dr. Rapalino: Postoperative MRI and MRA showed findings that would be expected after right encephaloduroarteriosynangiosis. There was a slight increase in the flow-related signal in the distal branches of the right middle cerebral artery that were near the surgical site and in the distal right M1 segment (Fig. 5).

Dr. Butler: After the brief hospitalization that followed the indirect revascularization procedure, the patient was discharged home with a prescription for daily aspirin and instructions to

maintain generous hydration, to minimize crying, and to contact his physicians if further transient neurologic deficits occurred. Follow-up MRA was scheduled for 5 months after hospital discharge to check for revascularization, as well as for evidence of ipsilateral recurrence and contralateral moyamoya disease.²¹ The patient was encouraged to resume his full range of activities but was advised to avoid intentional head collisions and to use a helmet for sports such as bicycling, skating, and skiing.

At 5 months of follow-up, the patient had had no further transient neurologic symptoms and was performing well in school. Follow-up MRA and PASL perfusion-weighted MRI showed revascularization in the right hemisphere. We discussed his specific risks associated with right external head trauma and reviewed activity guidelines and options for protective gear so that he may enjoy the benefits of vigorous exercise.

FINAL DIAGNOSIS

Moyamoya disease.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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